

Introduction to Cybersecurity

Fundamentals for Beginners — Theory Notes

These notes break down the core ideas behind cybersecurity into four parts: why digital security matters, the three foundational promises every secure system must keep, how attackers typically operate, and how defenders respond. Analogies and tables are used throughout to make abstract ideas concrete.

1. The Digital Landscape

The Reality of Data

The digital world runs on an invisible, constant flow of information. Every time you unlock your phone, buy something online, or send a message, you generate data that travels across networks and gets stored on servers somewhere.

What's at Stake

This data isn't abstract — it represents real assets: bank account access, private conversations, medical history, and identity documents. Because this information has real value, security exists to make sure only the right people can access or change it.

ANALOGY

Think of your personal data as the contents of your house — money, documents, photographs. Cybersecurity is simply the set of locks, alarms, and habits that keep that house safe from people who shouldn't get in.

2. The Foundations — The CIA Triad

Every secure system is built on three core promises, often remembered as the **CIA Triad**. Using the analogy of protecting a physical house: you want secrecy (only trusted people see what's inside), trustworthiness (nothing has been tampered with), and accessibility (you can get in whenever you need to).

Principle	What It Means	House Analogy
Confidentiality	Keeping secret information private — only authorized users can view files, read emails, or access accounts.	Curtains and locked doors so strangers can't see or enter your private rooms.
Integrity	Ensuring data stays accurate and unaltered, so systems remain reliable and trustworthy.	Knowing no one snuck in and rearranged your furniture or swapped your valuables for fakes.
Availability	Ensuring information and services are ready and fully functional whenever a legitimate user needs them.	Being able to open your own front door and get inside whenever you need to, without it being blocked.

NOTE

Most real-world attacks target one leg of this triad specifically — e.g. theft targets Confidentiality, tampering targets Integrity, and DDoS attacks target Availability. Recognizing which leg is under threat helps defenders respond correctly.

3. The Attacker's Playbook

Casing the Joint

Cyberattacks usually follow a systematic process rather than a random guess. It typically starts with **reconnaissance** — quietly gathering information about a target — followed by **scanning** for vulnerabilities in their systems.

The Easiest Entry Point

Attackers generally look for efficient shortcuts rather than brute-forcing the strongest defenses head-on. This is the **"weakest link"** principle: a system is only as secure as its most poorly protected component — often a careless human, rather than the technology itself.

Common Threat Vectors

Vector	How It Works	CIA Leg Affected
Phishing	Tricks human users into entering credentials on fraudulent websites or messages disguised as legitimate sources.	Confidentiality
SQL Injection	Malicious code is inserted into a website's input fields, confusing the underlying software into dumping its entire database.	Confidentiality / Integrity
DDoS (Distributed Denial of Service)	A botnet — an army of infected computers — floods a platform with traffic to overwhelm and disrupt it.	Availability

KEY TAKEAWAY

Most successful attacks don't rely on advanced hacking skill — they exploit a single overlooked weakness, whether that's an unpatched server, a careless click, or a poorly validated input field.

4. The Defender's Response Strategy

The Defender's Dilemma

Security is asymmetric: an attacker only needs to find **one** mistake to break in, while defenders must continuously manage risk across an entire attack surface — every device, account, and piece of software the organization owns.

Core Tooling

- **Firewalls:** act as structural boundaries that filter traffic entering or leaving a network.
- **Intrusion Detection Systems:** function like alarms, flagging suspicious activity as it happens.
- **Patch Management:** continuous software updates that close known vulnerabilities before attackers can exploit them.

Multi-Factor Authentication (MFA)

Passwords alone are no longer considered sufficient. Security teams instead validate identity by combining evidence from multiple distinct factor types:

Factor Type	Description	Example
Something you know	Information only the user should know.	A standard password or PIN
Something you have	A physical item the user possesses.	A one-time code sent to a mobile phone
Something you are	A unique physical trait of the user.	Fingerprint or face biometric scan

TIP

MFA is powerful because it requires an attacker to compromise multiple, unrelated factors at once — stealing a password is no longer enough on its own to break in.

✓ Quick Recap

Concept	One-Line Summary
CIA Triad	Confidentiality, Integrity, and Availability — the three promises security must keep.
Attack Pattern	Reconnaissance → Scanning → Exploiting the weakest link.
Common Vectors	Phishing, SQL Injection, and DDoS attacks.
Defender's Dilemma	Attackers need one flaw; defenders must protect everything.
MFA	Combines 'something you know,' 'have,' and 'are' to verify identity.

Source: based on "Introduction to Cybersecurity | Fundamentals | From beginner level" video notes.